

Ivonne Berenice Lemus Martínez | Embedded Software Engineer

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Embedded Software Engineer with over 5 years of experience in designing, developing, and executing complex embedded software projects. Proficient in **C** and **C++**, with proven expertise in applying **AUTOSAR** architecture across the automotive software development lifecycle. Skilled in **debugging** and testing embedded systems to identify and resolve issues efficiently. Experienced in real-time operating systems (**RTOS**) and low-level programming. Strong background in version control tools such as Git and Plastic, along with in-depth knowledge of communication protocols including **CAN**, **I²C**, **SPI**, and **UART**.

PROFESSIONAL EXPERIENCE

Embedded Software Engineer - BorgWarner. 19-03-2024 – 09-06-2024

- Integrated **pull request** code updates into the project's main branch using **version control tools** such as **Bitbucket**, **Git**, **GitHub**, **Plastic**, and **SmartGit**.
- Collaborated with a **global cross-functional team** to develop customized solutions that resolved customer needs.

Embedded Software Engineer - CIDECE. 16 - 06 2021 - 04-12- 2023

- Designed and implemented a **state machine** in **C** to control the horn feature output based on CAN signal inputs within the Electronic Control Unit of an automotive hardware system.
- Developed a **C software** component within the **AUTOSAR** architecture to monitor the ECU circuit board temperature.
- Achieved an 80% improvement in **C-language** software development efficiency through the precise specification of **software requirements** in accordance with the **Agile methodology**.
- Improved **ECU** reliability by resolving over 10 code **issues** using analysis and **debugging** tools such as **Ozone** and the **JTAG** interface to identify root causes.
- Completed tasks on schedule, including software design testing, **software requirements** analysis, and **peer code reviews**, using **Jira** and **Polarion** management tools.
- Implemented **software development** in accordance with the V-Model of the **ASPICE (Automotive SPICE)** process to fulfill the Software Development Life Cycle.
- Conducted **cybersecurity** testing to ensure the correct generation and secure storage of **cryptographic** keys analysis for **safety-critical** systems.
- Designed software **architecture** using dynamic and activity **UML** diagrams in **Enterprise Architect**, increasing coding efficiency by 80%.
- Identified 20+ code vulnerabilities through static analysis using LDRA and TRICORE tools in Eclipse, reporting MISRA violations, **Secure C** issues, and quality warnings.
- Performed software **unit tests** with VectorCAST to verify the detailed software design, ensuring high-quality software delivery and full condition coverage of functions.

Project Engineer - CIDESI. 12-04-2019 -14-06-2021

- Developed **low-level drivers** for **I²C**, **CAN**, **SPI**, & **UART** communication protocols, supporting **debugging** and validation activities, including **troubleshooting** issues.
- Developed software in **Embedded C** language for **ARM Cortex microcontrollers** to store data in external NAND flash memory, improving storage speed by 80% in a glider datalogger system.
- Designed and implemented **C** software using **SafeRTOS**, adhering to **DO-178** considerations for E Development Assurance Level, to control the speed of an inspection robot for a fuel jet pipeline.

- Enhanced the Mexico City railway software by implementing control **algorithms** in **SafeRTOS** and **FreeRTOS** to manage the speed and acceleration of the railway driving system, resulting in up to a 90% increase in system reliability.
- Developed embedded software in **C++** to create an interface for robot control, enabling abstraction and modularity through organized access to microcontroller peripherals.
- Performing **software testing** on train tracks and troubleshooting during system start-up to verify correct implementation of the system update.
- Performed reverse engineering on more than 20 analog amplifier circuit **PCBs** by analyzing their design and validating signals using an **oscilloscope**.
- Implemented **C++ software** using Object-Oriented Programming (**OOP**) to control the speed and acceleration of Mexico City's railway driving system, enabling project flexibility and scalability.
- Maximized the efficiency of TMS570LS3137 **microcontroller** system up to 30%, by developing **embedded software** implementing **RTOS (Real Time Operating System)**, using **GIT** for program version control.
- Visited customer and internal locations to provide hands-on support to **systems engineers**, identify **hardware** requirements, and contribute to the digitalization of the existing railway driving system in Mexico City.
- Enhanced the Mexico City railway software system by applying **MISRA-C** and **Secure-C** coding standards, resulting in up to a 90% increase in peripheral **communication security** and reliability.

Graduate Studies- Instrumented equipment Department - CIDESI. 27-09-2017 – 15-12-2020

- Created an electronic **architecture** for signal acquisition of 80 Hall sensors implemented in **FPGA**, which allowed to get 400 μ s in acquisition time and reduce by 70% the implementation costs.
- Implemented **I²C** and **SPI** communication protocols on an embedded system using **C language** to transmit and store acquisition data from Hall sensors.
- Assisted in prototyping a mechatronic device for gas pipeline inspection validating the Magnetic Flux Leakage (MFL) technique.

Internship - Instrumentation Equipment Department - CIDESI. 27- 04-2017 - 15-09-2017

B.S. in Mechatronics Engineering | 09 -02-2012 - 10-02-2017

National Technological Institute of Mexico | San Luis Potosí

- Developed a virtual instrumentation applying least squares **algorithms** to model the dynamic control of a DC motor using **Embedded C** language on a **TI microcontroller**.

EDUCATION

B.S. in Mechatronics Engineering | 09 -02-2012 - 10-02-2017

National Technological Institute of Mexico | San Luis Potosí

- Developed **firmware** for Texas Instruments, Microchip, and ST **microcontrollers** with **Assembly** and **C languages**.

M.Sc. & T. Mechatronics | 07-09-2017 - 09-12-2019

CIDESI | Querétaro

- Developed and executed **embedded** software to test a **hardware architecture** for analog data acquisition.

SKILLS

Programing Languages: C, C++ & Assembler

DevOps: Git & Plastic

CPU architectures: FPGA (Artix-7), FPGA (Spartan 3E), TMS570LS3137, MSP430F2618, PIC4550, STL microcontroller.

Communication protocols: I2C, SPI, UART & CAN

Operating systems: RTOS, SafeRTOS & FreeRTOS

Languages: English & Spanish

COURSES

C++: UDEMY. 2024-09-10 - 2024-10-10

Vector tools: CIDECE. 2021-06-16 - 2021-09-27